

SETTING Up CI/CD

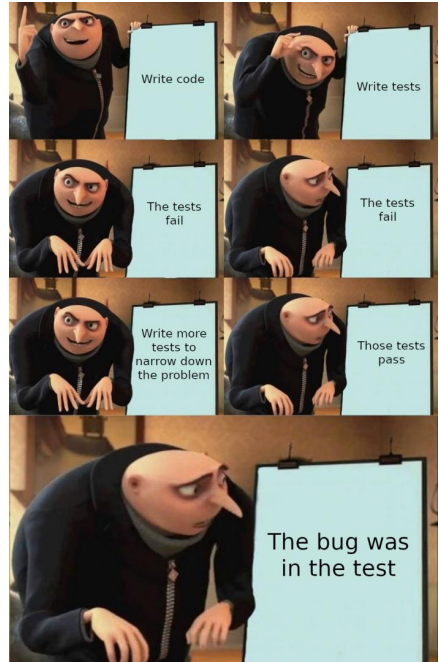
Oct 07, 2026



Brenan Keller
@brenankeller

A QA engineer walks into a bar.
Orders a beer. Orders 0 beers.
Orders 999999999999 beers.
Orders a lizard. Orders -1 beers.
Orders a ueicbksjdhd.

First real customer walks in and asks where the bathroom is. The bar bursts into flames, killing everyone.



Write tests

The tests fail

The tests fail

Write more tests to narrow down the problem

Those tests
pass

The bug was
in the test

AGENDA

① TESTS!! Wait what??
What's that?!

② What is CI/CD



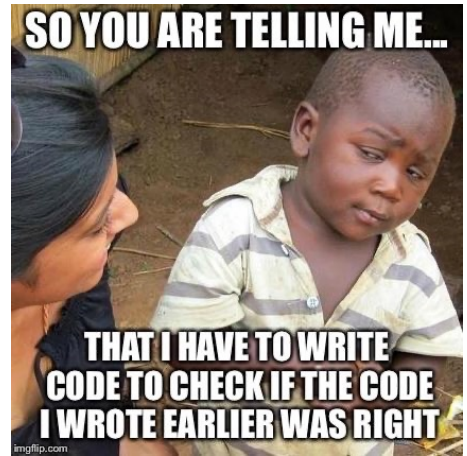
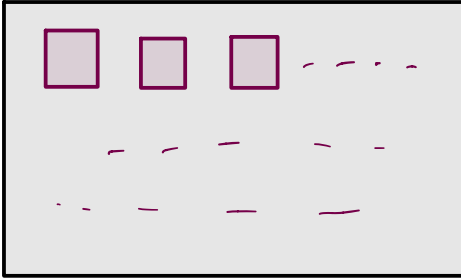
Dave Dustin
@venzann

Genie: I'll give you one billion dollars if you can spend 100M in a month. There are 3 rules: No gifting, no gambling, no throwing it away.

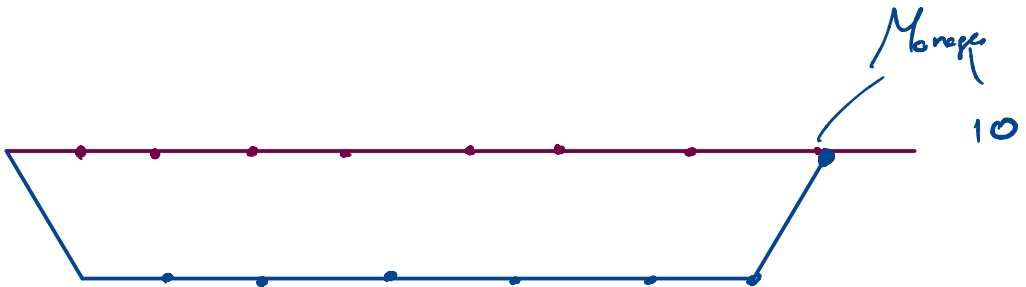
SRE: Can I use AWS?

Genie: There are 4 rules.

TESTS



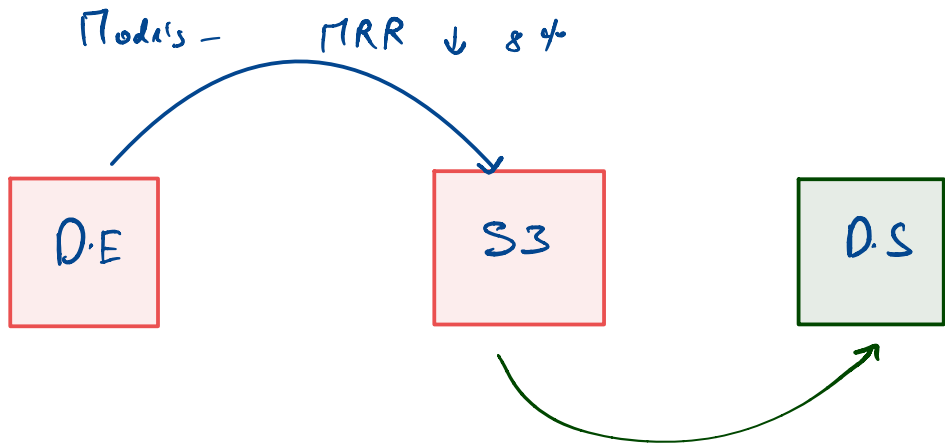
if you want to deploy a new feature, it takes about 1 year to pass all the relevant tests, from all the relevant teams.



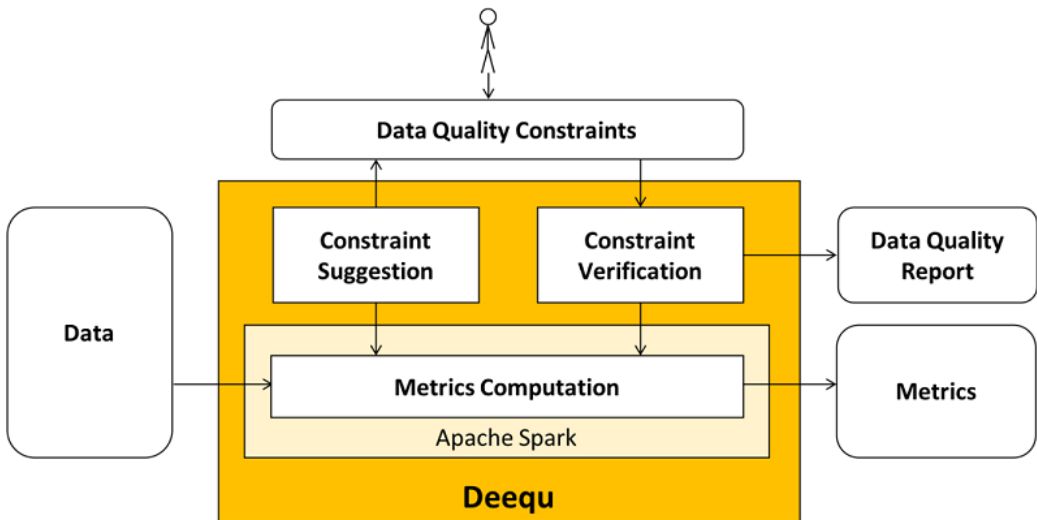
What are some core objectives for testing?

1. Entire code-base should be "functional".
2. Any online/offline metric shouldn't be hampered.
3. Latency should not increase/decreases in an unexpected way.
4. Data Integrity should not be compromised.

Latency = Time taken for my request to arrive at the remote server
+
Time taken for the model to generate a response.
+
Time taken for the response to arrive at my computer.



<https://aws.amazon.com/blogs/big-data/test-data-quality-at-scale-with-deequ/>



Recommended alternatives for amazon deequ for data testing

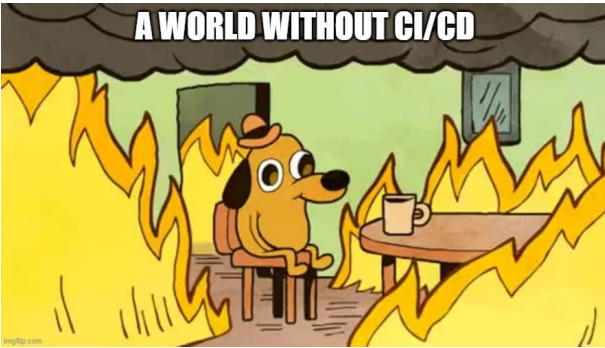
Worked for 25s

Yes. The strongest alternatives to **Amazon Deequ** today are **Soda Core**, **Great Expectations**, **Pandera**, and **dbt tests**, but they target somewhat different environments.

One important point first: **Deequ itself is not abandoned**. The project is actively maintained; the latest release is 2.1.0, and it continues to target Spark 3.x workloads. So I wouldn't migrate purely because you're worried about maintenance. [GitHub +1](#)

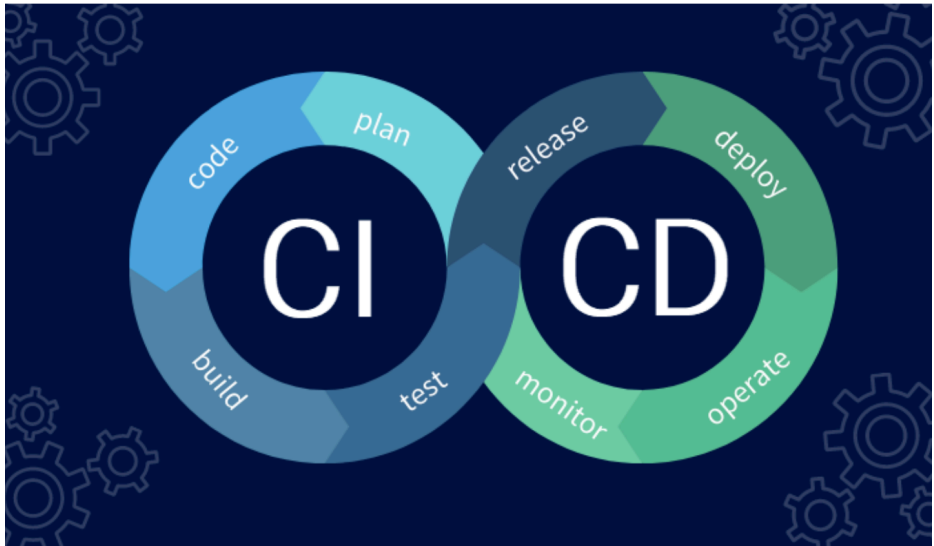
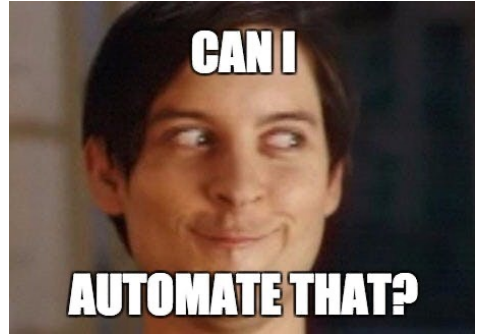
CI/CD

A WORLD WITHOUT CI/CD



CAN I

AUTOMATE THAT?



```
import pytest
import numpy as np

from hello_flask import pancakes
```



— Connection
Process of
establishing
Connection

purpose of fixture is to create resources that can be shared across tests

```
# pytest fixture for connecting to a database
@pytest.fixture
def db_connection():
    # # Simulate a database connection
    # connection = "Database connection established"
    # yield connection
    # # Teardown code
    # connection = None
    # print("Database connection closed")

## use the fixture in a test
def test_db_connection(db_connection):
    # assert db_connection == "Database connection established"
```



```
@pytest.fixture
def client():
    return pancakes.test_client() # test_client() is a function provided by Flask to simulate
    requests to the application without running a server
```

▷ Generate Mermaid Diagram | 🗨 Open Chat @mermaid-chart